# CardioTwin

## Coronary risk with stated uncertainty, and echo education

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Research prototype. Not a medical device. MIT License.

# Problem statement

Coronary artery disease is one of the leading causes of death worldwide. Whether a patient with chest pain or breathlessness needs invasive angiography depends on routine findings that often do not point clearly one way.

- Simple risk scores return one number, often without uncertainty or vessel-level detail.
- Trainees have few chances to match echocardiogram views to cardiac anatomy.
- CardioTwin asks whether an open, transparent tool can help with both gaps.

It supports clinical reasoning. It does not replace angiography or clinical judgment.

# Healthcare use case (from the codebase)

- Vessel-level risk from 54 routine inputs: demographics, risk factors, symptoms, ECG, echo and labs.
- Transparent output: bootstrap intervals, calibration method and recorded limitations.
- Hash-sealed evidence passports for traceability.
- Echo trainer: 3D heart, six probe positions and a synthetic echo (simulation).
- Six echo calculators that return raw results with sources. No clinical thresholds applied.
- MedClear: plain-language clinical notes, offline. Pulse 911: emergency dispatch simulation.

Intended users: cardiology clinicians and trainees, and researchers. Not intended for patient-care decisions.

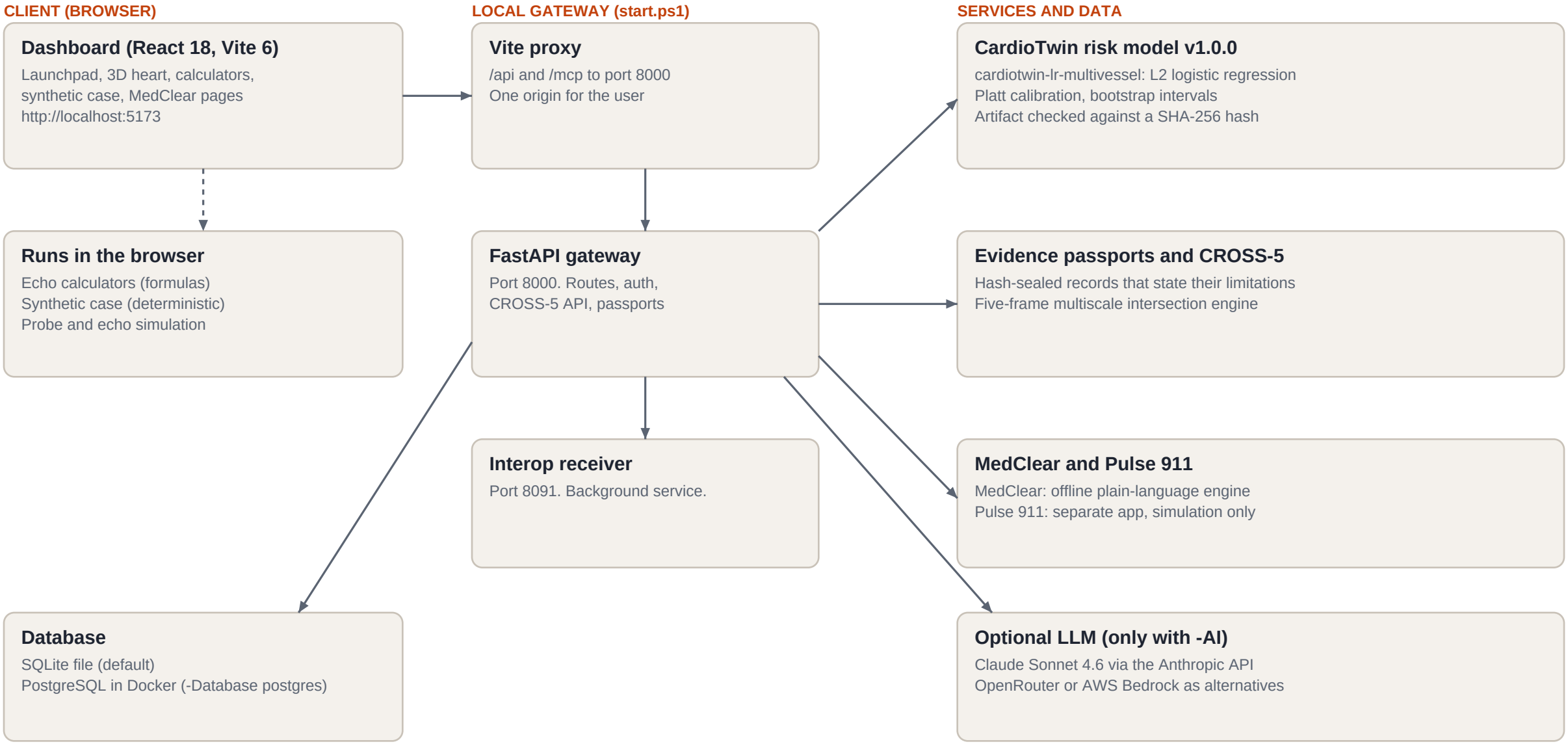
# What the demo shows

| Route | What it does | Status |
|---|---|---|
| /platform | Launchpad for every section | Working |
| /cardiotwin | 3D torso, heart and probe, synchronised synthetic echo | Working. Simulation. |
| /cardiotwin/calculators | Six echo calculators with worked examples | Working |
| /cardiotwin/case | Evidence-linked synthetic case, what-if, CROSS-5 status | Working. Synthetic data. |
| /cardiotwin/risk | CAD risk prediction, sensitivity and model trust | Working |
| /medclear | Plain-language summaries of clinical notes | Working |
| /pulse | Pulse 911 emergency copilot studio | Simulation |
| /cross5 | CROSS-5 multiscale intersection engine | Experimental |

# CardioTwin architecture

One local origin for the user. start.ps1 launches every service behind it.

## Architecture




Only port 5173 is opened by the user. Ports 8000 and 8091 run in the background, and the calculators and synthetic case run in the browser.

# Technical stack

| Layer | Technology |
|---|---|
| Risk model | Python 3.12, scikit-learn logistic regression, NumPy |
| Backend API | FastAPI 0.115, Uvicorn 0.32, Pydantic 2.9 |
| Dashboard | React 18.3, TypeScript 5, Vite 6, React Router 7 |
| 3D | Three.js 0.169, React Three Fiber 8.17, Drei 9.114 |
| Pulse 911 app | React 19, Vite, Tailwind CSS v4 |
| MedClear | Python 3.12 |
| Tests | pytest (backend), Vitest 4 (dashboard) |
| Data stores | SQLite (default), PostgreSQL in Docker (optional) |
| Launcher | PowerShell (start.ps1) |
| Optional AI | Anthropic SDK 0.39 (Claude); OpenRouter and AWS Bedrock as alternatives |

# AI/ML model: cardiotwin-lr-multivessel v1.0.0

- One L2-regularised logistic regression each for CAD, LAD, LCX and RCA. Regularisation chosen by inner 5-fold CV.
- The shipped CAD output is max-projected, so P(CAD) is at least the largest vessel probability.
- Platt calibration adopted for all four outputs, under a rule requiring the Brier gain to exceed its standard error.
- Intervals from 1,000 patient-level bootstrap resamples. No conformal coverage guarantee for the shipped artifact.
- Training data: Z-Alizadeh Sani extension (UCI 411), 303 patients, CC BY 4.0, single centre in Iran.
- LAD, LCX, RCA and catheterisation are excluded from the CAD inputs to prevent label leakage.
- The risk model does not use an LLM. Claude Sonnet 4.6 is optional and runs only with -AI.

# Outcomes: model results (out-of-fold)

| Output | ROC-AUC | 95% CI (patient-level bootstrap) |
|---|---|---|
| CAD, direct model | 0.913 | 0.876 to 0.944 |
| CAD, shipped (max-projected) | 0.912 | not reported separately |
| LAD | 0.823 | 0.776 to 0.868 |
| LCX | 0.679 | 0.615 to 0.734 |
| RCA | 0.738 | 0.682 to 0.788 |

CAD direct model: PR-AUC 0.961 (0.938 to 0.979); Brier 0.103 (0.081 to 0.125); ECE 0.054. Protocol: 3 x 5 repeated stratified cross-validation.

These come from one centre with no external validation. Discrimination for LCX and RCA is modest, and performance in other populations is unknown.

# Outcomes: engineering verification

| Check | Result (last recorded run) |
|---|---|
| Dashboard type check | 0 errors |
| Dashboard unit tests | 439 of 460 pass. 21 fail in 7 files; these failures predate this work. |
| Probe geometry and echo simulation | 36 tests pass |
| Calculator formulas and registry | 42 tests pass |
| Evidence-linked case model | 15 tests pass |
| Backend CROSS-5 tests | Pass (subset run with -k cross5) |
| MedClear | 2 of 2 pass |
| Not yet run | Full backend pytest; 390 px and 768 px layout checks; production build after final navigation edits; launcher re-run |

# Limitations and safety

- Research prototype. Not a medical device. Not CE-marked or FDA-cleared.
- Single-centre training data. No external validation yet.
- Intervals describe sampling uncertainty, not performance in other hospitals.
- Outputs are not diagnoses. Clinical confirmation, such as angiography or CT, is required.
- Calculators give raw values only. Pulse 911 is a simulation, not a validated triage system.
- The 3D heart and echo simulation are educational, and their two clocks are not frame-locked.

# Open source and third-party licences

- Original work: MIT License, Copyright (c) 2026 V. S. Rupesh Kumar (LICENSE).
- Eight module folders keep their own licence files: MIT, BSD 3-Clause and EPL 2.0.
- Third-party data and assets: UCI 411 (CC BY 4.0), BodyParts3D-derived mesh (CC BY-SA 2.1 Japan), HRA heart model (CC BY 4.0).
- 23 module folders have no licence text, so this licence does not cover them. Their status is open (NOTICE.md lists them).

# Demo video and next steps

**Demo video: PENDING. A 15 to 20 minute unlisted YouTube video must be recorded and uploaded. Its link goes in README.md under Demo video.**

- Proposed: validate the model on an independent cohort before any clinical claim.
- Proposed: clinician review of the outputs, the uncertainty display and the wording.
- Proposed: finish the full backend test run and the layout checks at phone and tablet widths.
- Proposed: resolve the licence status of the 23 module folders and the adapted code in core.